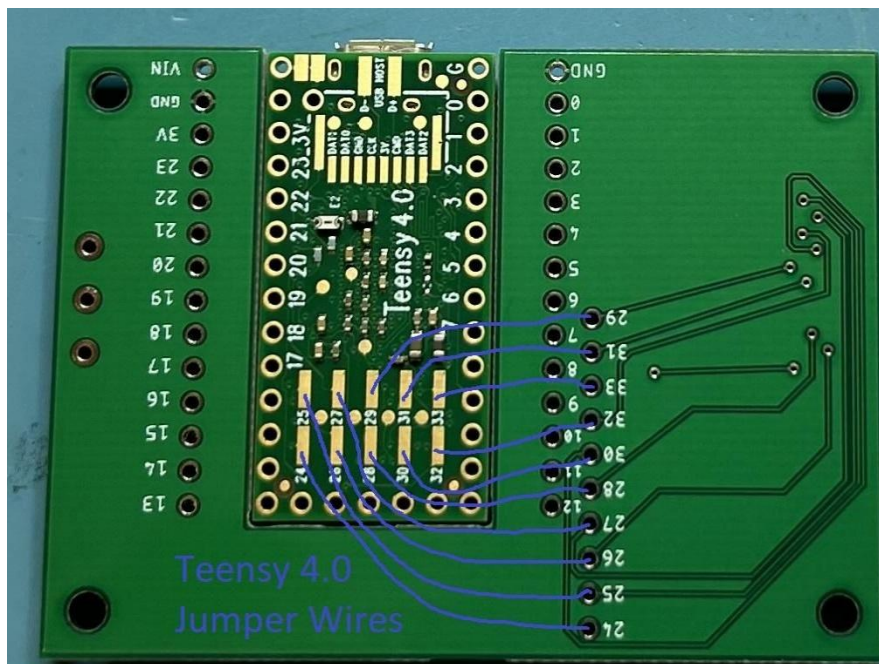


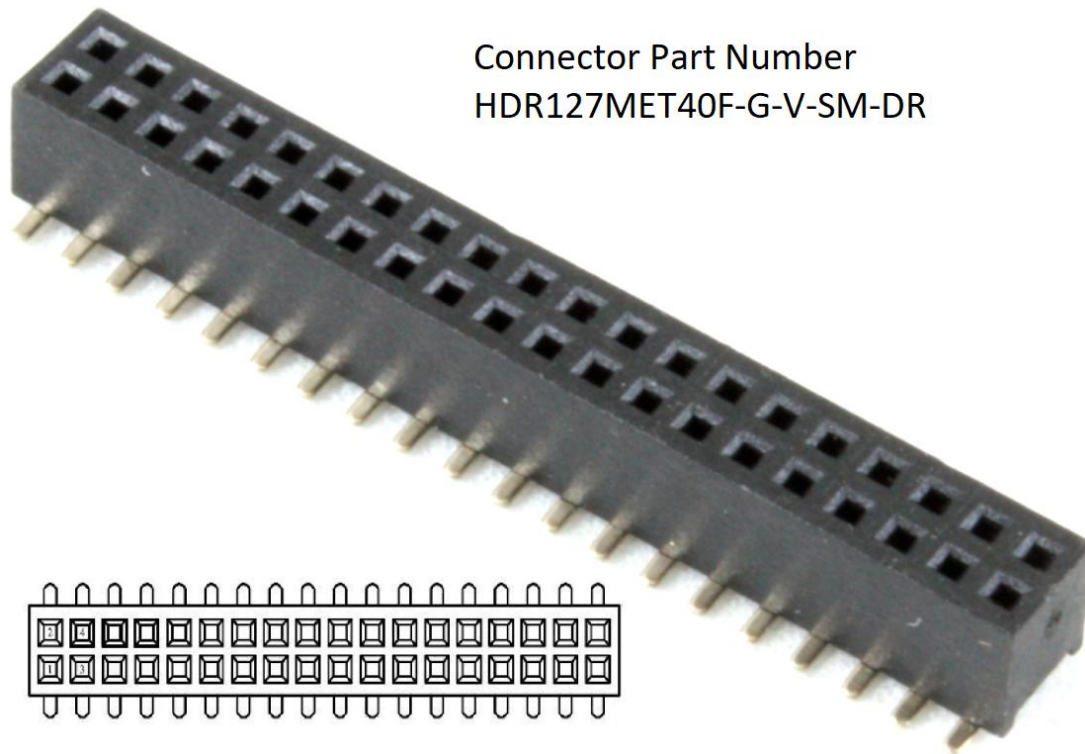


## Apple iBook G3 Clamshell keyboard with 40 pin connector

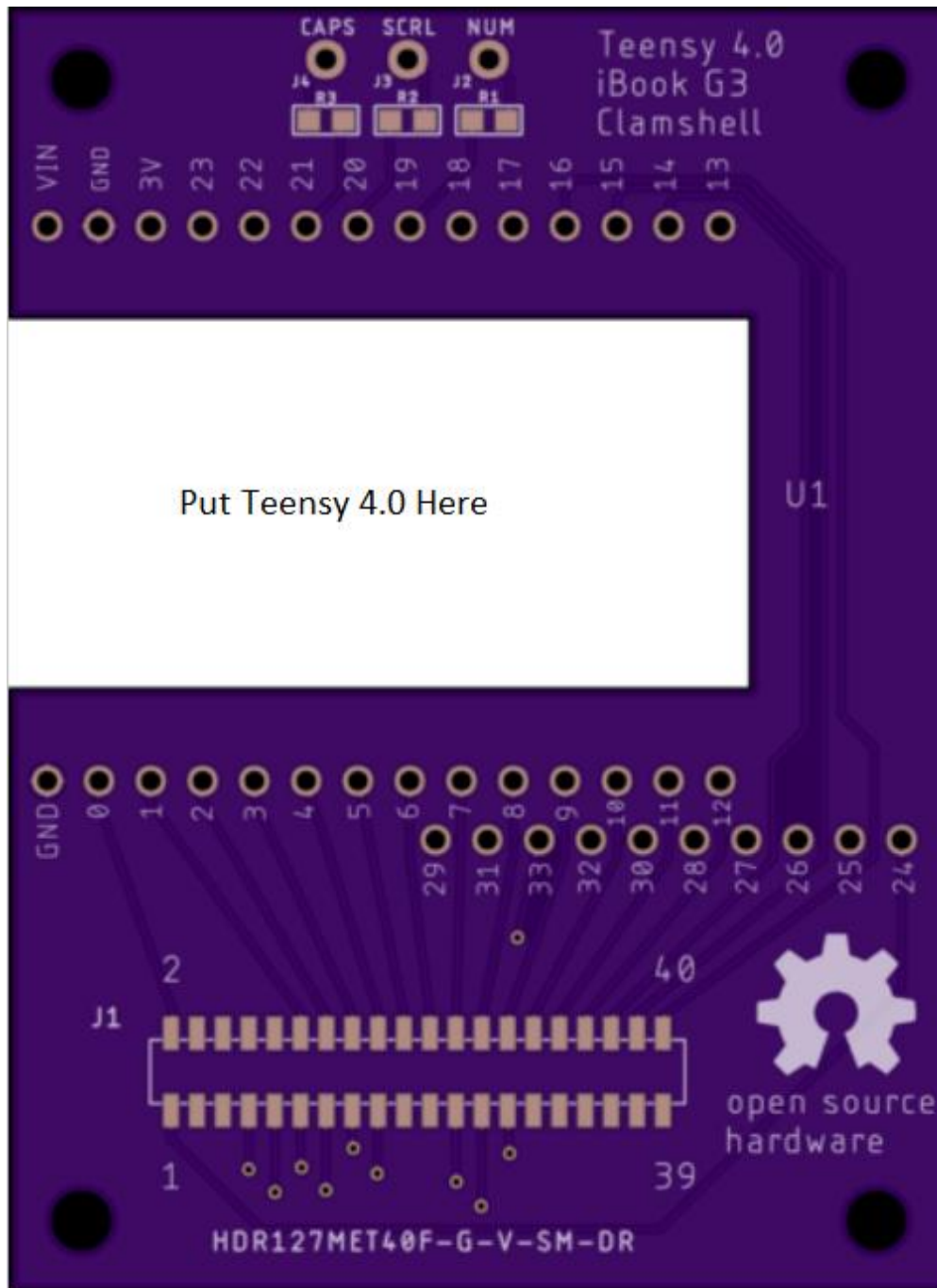
"Billy67" has used my code and circuit boards to convert several Apple keyboards to USB. Many are documented in the "I made it" section at the end of my Instructable. One of those keyboards is from an iBook G3 clamshell which has a special 40 pin connector on the end of its FPC cable. Because of height restrictions in the clamshell laptop, the Teensy must sit inside a cutout in the board. Wires or "U" shaped header pins connect Teensy I/O's 0 through 23 to the connector board. I/O's 24 through 33 use backside wires as shown below.



Billy did some experimenting and found connector part number [HDR127MET40F-G-V-SM-DR](#) mates up with this keyboard cable perfectly.



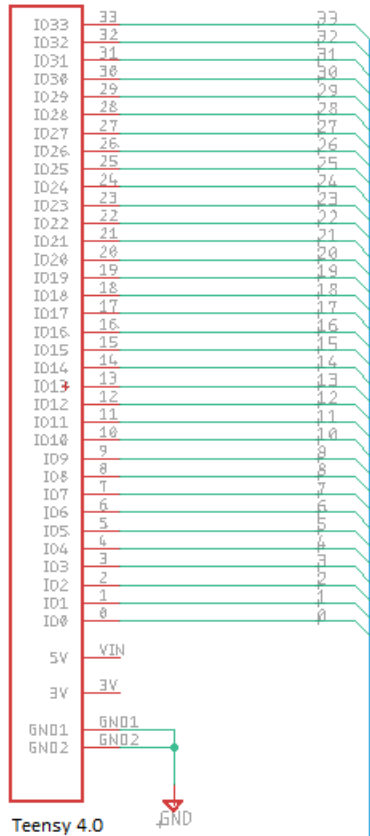
To build a prototype, he etched his own circuit board with fine pitch traces drawn freehand with an ink pen (wow, that's hard core). If you're like me and haven't used PCB etchant in a few decades, you're going to want a circuit board file to send to a fab house. I originally used a Teensy 3.2 but have updated the board design to use a Teensy 4.0. The new Eagle layout file is "iBookG3Clamshell\_T4p0.brd". The zip'ed Gerber file is "iBookG3Clamshell\_T4p0\_2023-10-31.zip".



Eagle file iBookG3Clamshell\_T4p0.brd

Gerber file iBookG3Clamshell\_T4p0\_2023-10-31.zip

These files and the keyboard code are available at my [repo](#). Teensy I/O 13 is not used so the LED can remain on the board. I/O's 19, 20, and 21 can be used for NUM, SCROLL, and CAPS lock LEDs with current limit resistors. The following schematic shows the connections from the Teensy to the keyboard connector.



## IBook G3 Clamshell

